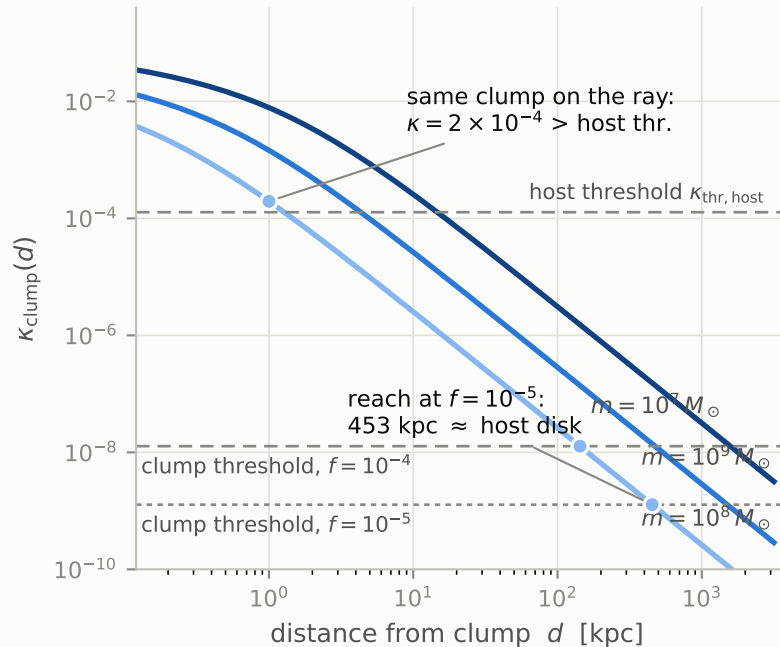
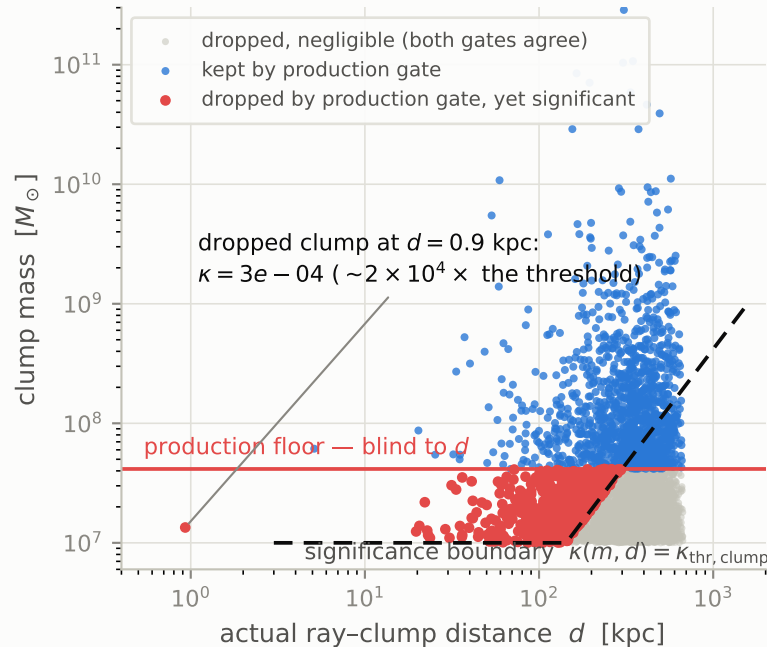


A — one clump: $\kappa \propto m/d^2$ spans 5 decades, so the threshold must price the worst-case d



B — matched catalog, ray at $r = 300$ kpc, $f = 10^{-4}$: the r -keyed gate discards 61% of this catalog's clump κ^2



C — the observable: population substructure κ -variance bias vs brute force

